Training-Free Cultural Alignment of Large Language Models via Persona Disagreement

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Abstract

Large language models are increasingly deployed in systems that influence morally consequential decisions, from content moderation to clinical prioritization, yet they make moral judgments as if the whole world thinks like the West. The Moral Machine experiment [Awad et al., 2018] confirmed this is wrong at scale: 40 million judgments across 233 countries reveal that moral preferences are systematically structured by culture. A model that ignores this variation does not simply underperform but imposes one society's moral intuitions on all others. We show that within-country *demographic disagreement*, not consensus, provides a sufficient signal to close this gap at inference time, without modifying weights, requiring per-country labels, or accessing internal activations, the only regime viable for global deployment where hundreds of cultures must be served through black-box APIs. Our method, **DISCA** (Disagreement-Informed Steering for Cultural Alignment), instantiates each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2022]; their disagreement on a moral dilemma pinpoints where the base model is miscalibrated and by how much, while agreement indicates no correction is needed. The method is self-regulating: corrections are bounded by a fairness-aware aggregation that prevents any demographic group from being sacrificed, and shrink toward zero when the underlying signal is noisy or unreliable. Across 20 countries, four continents, and six model families (3.8B–70B parameters), the method reduces cultural misalignment by 19–24% on the MultiTP benchmark [Jin et al., 2025], with gains spanning geographies rather than concentrating in any single region. Ablations confirm that persona diversity and positional debiasing are the dominant components, and that failure is architecturally predictable, not culturally systematic.

1 Introduction

Large language models are increasingly deployed in systems that touch morally consequential decisions, from content moderation to clinical prioritisation to risk assessment, and they serve users across dozens of countries and languages. Yet when these models face moral dilemmas, their responses fail to reflect the cultural diversity of the populations they serve [Jin et al., 2025]. They do vary across languages, but the variation does not track documented human moral references [Awad et al., 2018]; it is culturally arbitrary rather than culturally informed. For instance, when a self-driving car must choose between its passengers and pedestrians, Japanese respondents overwhelmingly spare the pedestrians, ranking 1st out of 117 countries, while Chinese respondents show the opposite preference, ranking 116th [Awad et al., 2018]. No contemporary LLM reproduces this contrast when prompted in Japanese versus Chinese [Jin et al., 2025].

The scale of the mismatch is well documented. The Moral Machine experiment [Awad et al., 2018] collected 40 million judgments across 233 countries and found that moral preferences cluster into

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three broad cultural groups with systematically different priorities. On the age dimension alone, France sits near the top of he global ranking for sparing the young while Japan ranks 103rd and China 115th out of 117 countries. On pedestrians versus passengers, Japan ranks 1st and China 116th, a near-complete reversal between two countries often grouped together culturally. These differences are not noise: they are geographically structured, correlate with institutional and economic indicators, and are mediated by generational cohorts within each country [Awad et al., 2018, Inglehart and Welzel, 2005]. When the MultiTP benchmark [Jin et al., 2025] tested 19 LLMs on the same dilemmas in 107 languages, no model recovered this cultural structure. Per-language response clustering placed Persian with German and Swedish, and Portuguese with English and Chinese [Jin et al., 2025], groupings that bear no resemblance to the cultural geography documented by the Moral Machine. The most misaligned models went further, collapsing moral reasoning into extreme binary preferences: always spare women, always spare the young, always spare humans over animals, overriding the nuanced trade-offs that human respondents actually make [Jin et al., 2025].

Closing this gap is harder than it appears. Existing approaches each require at least one resource that is unavailable at the scale of global deployment. Fine-tuning or training culture-aware adapters [Zhang et al., 2026, Yao et al., 2025] produces models that reflect a target culture, but requires per-country training data and a separate model variant for every target culture. With 195 countries and substantial within-country variation, this mean we may need up to 195 models variants for this fine-tuning method. Thus, it does not scale. Reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] avoids weight updates but requires a reward model trained on moral preference data for each country, but country-specific moral annotations at this scale do not exist. Creating them would require extensive crowdsourcing campaigns in each country, raising practical questions about cost and coverage, as well as normative questions about who speaks for a culture's moral preferences. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires neither training data nor weight updates, but it requires access to the model's internal representations, which text-only API deployments do not expose. The simplest alternative, direct country-prompting ("Answer as a Japanese person"), avoids all three requirements but produces minimal improvement in practice [Jin et al., 2025]. Telling a model to "be Japanese" treats millions of people as if they share a single moral outlook, and the outlook the model produces tends to reflect training-data stereotypes rather than documented cultural values [Deshpande et al., 2023].A practical method must therefore update no weights, consume no per-country labels, and operate without access to model internals. To our knowledge, no existing method satisfies all three while providing empirically grounded cultural steering.

Our central observation is that the signal needed for cultural alignment already exists in publicly available survey data. Consider a dilemma that trades young lives against elderly lives, presented to four demographically grounded personas representing a single country: a young adult, a middle-aged professional, an older citizen, and a population-wide aggregate, each constructed from World Values Survey microdata [Haerpfer et al., 2022]. In a country like Japan, where all age cohorts agree that this trade-off is relatively balanced [Awad et al., 2018], the four personas produce similar responses, and the base model is typically already close to the cultural target. No correction is needed. In a country like France, where younger and older cohorts weigh youth differently [Awad et al., 2018, Inglehart and Welzel, 2005], the personas diverge, and the spread of their responses directly encodes both the direction of the needed correction and its magnitude. In other words, where personas agree, the model is already calibrated; where they disagree, the disagreement tells us what to fix and by how much. Therefore, within-country disagreement, not consensus, is the steering signal.

We operationalise this insight in a method we call **DISCA** (Disagreement-Informed Steering for Cultural Alignment), which proceeds in three steps. First, we represent each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2022]: three age cohorts (young, middle-aged, older) plus a population-wide aggregate. Each persona responds to moral dilemmas in the target country's native language, and the differences among their responses form the cultural signal. Second, we remove the positional bias that would otherwise mask this signal. When a moral dilemma is presented as "Option A: spare the young group / Option B: spare the elderly group," LLMs tend to favour Option A regardless of its content [Liu et al., 2024]. Swap the order, and the model favours the elderly group instead. This positional preference has nothing to do with culture, but it is strong enough to drown out the genuine cultural differences among personas. Removing it is not optional: until it is gone, the persona responses reflect the model's positional bias, not the country's moral values. We formalise this debiasing argument in the method and appendix derivation, and we argue it explains why prior prompting-based cultural steering has underperformed [Jin et al.,

2025]. Third, we combine the four personas' corrections into a single adjustment using a bounded, fairness-aware aggregation rule so that no demographic group is systematically sacrificed for the others. **DISCA** is self-regulating: when the signal is noisy or unreliable, corrections automatically shrink toward zero rather than committing to a potentially wrong adjustment.

Across 20 countries, four continents, and 6 open model families spanning 3.8B to 70B parameters, **DISCA** reduces cultural misalignment by 19–24% on the MultiTP benchmark [Jin et al., 2025] relative to vanilla decoding, with gains spanning the Americas, Europe, and Southeast Asia rather than concentrating in any single region. The lowest absolute misalignment comes not from the largest model but from Phi-4 (14B), suggesting that how well the model handles fine-grained preference distinctions matters more than raw parameter count.

Our contributions are as follows.

- We show that within-country demographic disagreement from public survey data provides a sufficient training-free cultural steering signal, and we introduce **DISCA**, the first inference-time alignment method satisfying all three deployment constraints simultaneously.
- We prove two results that constrain the method's design. First, positional framing bias makes debiasing a strict prerequisite for any persona-based steering: without it, the cultural signal is unrecoverable. Second, when combining corrections across demographic groups, loss-averse aggregation is the principled choice: it protects minority viewpoints that simple averaging would override, without fixating on the worst case alone.
- We validate **DISCA** across 20 countries and 6 model families, demonstrate that gains span geographies, and show that failure is predictable from base-model properties alone, enabling deployment-time risk assessment without running the full pipeline.

2 Related Work

Benchmarks and cultural evaluation. The Moral Machine experiment [Awad et al., 2018] established that moral preferences are culturally structured; MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs, providing the evaluation bed we adopt. Existing moral benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric, and broader diversity efforts [Kirk et al., 2024, Durmus et al., 2023, Qiu et al., 2025] cover fewer languages or different modalities. We validate on MultiTP's binary trolley vignettes where AMCE vectors provide a clean six-dimensional surface; a pilot extension to the open-ended BLEnD benchmark [Myung et al., 2024] (Appendix A15) reveals that the binary logit-gap formulation does not transfer directly to free-form generation, and no existing training-free method has demonstrated country-level calibration on such tasks. On the diagnostic side, LLM outputs correlate most strongly with WEIRD respondents [Santurkar et al., 2023], RLHF introduces unintended cultural shifts [Ryan et al., 2024], and recent work shows that LLMs systematically stereotype the moral values of non-Western populations [Zewail et al., 2026]-underscoring the practical need for culturally grounded correction mechanisms.

Inference-time alignment. The pluralistic alignment vision [Sorensen et al., 2024] argues that no single preference ordering serves all communities; inference-time methods are a natural vehicle because they avoid baking a fixed value stance into weights. However, existing approaches each sacrifice one of the three constraints we target: activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access; reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] needs per-country reward models; and culture-aware adapters [Zhang et al., 2026, Yao et al., 2025] need per-culture training. Recent logit-level steering work [An et al., 2026] and culturally localised generation via sparse autoencoders [Khanuja et al., 2026] demonstrate that the logit surface is a richer control interface than previously appreciated, but neither targets country-conditioned moral calibration. Inference-time stochasticity can improve moral alignment by reshaping confidence [Kwon et al., 2026]; DISCA complements this by targeting *directional*, survey-grounded corrections rather than generic uncertainty inflation, and we include it as an explicit baseline (Appendix A17). Black-box alignment modules that learn small guidance policies from preference data [Bobbili et al., 2025, Chen et al., 2025] sit in an adjacent regime: they operate on frozen model outputs, but still require a training stage for the guidance policy and are typically defined over sentence-level continuations rather than binary decisions. Adaptive importance sampling in pre-logit space [Kanai et al., 2025] achieves

similar variance control but requires white-box activation access; we work strictly with decision-token logits. Training-time prospect-theoretic alignment [Ethayarajh et al., 2024] extends KL-regularised preference learning with loss-aversion via a Kahneman–Tversky value function; DISCA applies the same value shape at *test time*, per scenario, so the asymmetry aggregates heterogeneous personas rather than shaping a trained policy. To the best of our knowledge, DISCA is the first training-free method targeting country-level moral calibration specifically.

Theoretical grounding. DISCA sits at the intersection of three theoretical traditions, each supplying one ingredient that the others lack. *Control as probabilistic inference* [Levine, 2018, Todorov, 2009] recasts optimal decision-making as posterior inference over an intention distribution; path-integral steering [Williams et al., 2018] instantiates that view with softmax-weighted perturbations that provably approximate the KL-regularised optimal control under mild regularity conditions [Todorov, 2009]. This is what licences us to read our Eq. 5 as an inference step rather than an arbitrary scalar interpolation, and it motivates the Gaussian proposal around $\bar{\delta}$ as the entropy-maximal default in the absence of stronger priors. *Prospect Theory* [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] then supplies the utility shape: the key empirical regularity-that human decision makers weight a loss of magnitude x roughly $\kappa{\approx}2.25$ times more than a gain of the same magnitude-is exactly the asymmetry we want an LLM cultural controller to respect, because a correction that silences one demographic persona is worse than a correction that fails to help it. Recent evidence that prospect-theoretic biases emerge naturally in LLM decision-making under risk [Payne, 2025] strengthens the match: the asymmetric gain/loss structure we impose *mirrors* one the models themselves already exhibit, rather than introducing an alien objective. *Importance-sampling (IS) theory* [Elvira and Martino, 2021] finally supplies the variance-control machinery, and it is here that self-normalised IS is known to be biased at finite K; the dual-pass construction we adopt is a light-weight bootstrap-style variance probe [Efron, 1979] that uses the same compute budget to detect instability rather than silently accept it, paired with an effective-sample-size guard grounded in design-effect theory [Kish, 1965]. We treat the dual-pass gate as a variance heuristic rather than a formal unbiasedness certificate; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A4).

3 Method: DISCA

The pipeline has three logical stages: extract cultural signal from a WVS-grounded persona panel (§3.2–§3.4), convert dispersion into a bounded correction via Prospect-Theory importance sampling (IS) with dual-pass reliability (§3.5), and stabilise across scenarios via a hierarchical country prior (§3.6). Algorithm 1 ties the three together; rationale for each choice is summarised in Appendix A1.

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` (a) and `B` (b). For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_a,\, z_b]$ at the decision token position; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_b - z_a$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention [Jin et al., 2025]. All six dimensions-including Utilitarianism-are computed uniformly as the empirical mean of p_{spare} (Appendix A2).

Goal: Without modifying θ, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$. We primarily measure alignment with the *misalignment score* (MIS): ℓ_2 distance between $\hat{\mathbf{m}}^{(c)}$ and $\mathbf{h}^{(c)}$ (lower is better). Jensen–Shannon divergence and Pearson r are reported as secondary diagnostics where noted.

3.2 Cultural Persona Construction

We use $N{=}4$ personas-three age cohorts plus a country-wide aggregate-grounded in WVS Wave 7 individual-level microdata [Haerpfer et al., 2022]. The choice of $N{=}4$ is the smallest panel that

covers the primary demographic axis of within-country variance while preserving a population anchor against cohort-specific noise [Inglehart and Welzel, 2005, Kish, 1965]; full justification is in Appendix A1.

For each country c, following the 10-variable cultural-value scheme of Greco et al. [2026], we extract: *religiosity* (Q6P), *child-rearing values* (Q17P), *moral acceptability* (Q177–Q182), *social trust* (Q57P), *political participation* (Q199P, Q200P), *national pride* (Q254P), *happiness* (Q46P), *gender equality* (Q29P–Q33P), *materialism orientation* (Q152–Q153), and *tolerance for diversity* (Q19P–Q23P). Items from the "inverted" WVS-7 CSV (suffix P) are directionally coded so that higher values consistently indicate the positive pole of each dimension. For each country, respondents are segmented into three age cohorts-*young* (<36), *middle-aged* (36–55), and *older* (>55)-yielding three cohort-specific personas that capture generational variation in cultural values. A fourth persona aggregates all respondents regardless of age, anchoring the ensemble with the country's population-wide cultural centre of gravity. All four personas are thus data-driven and country-specific; no agent carries a hard-coded moral stance that could bias particular dimensions. Dimension means are normalised to $[0, 1]$ and mapped to four-level natural-language descriptors via quartile cuts (≥ 0.75, ≥ 0.50, ≥ 0.25, < 0.25); see Appendix A3. Each persona is a native-language system prompt assembled from a header, one sentence per WVS dimension with data, and a moral-reasoning closing line.

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix A3.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} scale logits before computing decision gaps. We use $T_{\text{cat}} = 4.0$ (Species), 3.5 (Gender), and 1.5 for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same values appear in Appendix A5.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here δ_{base} is the decision gap of the base prompt (no persona). The reward r_i measures how strongly persona i shifts the model *away from the culturally-neutral baseline*: a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit. (Note: the scalar r introduced later in Eq. 6 denotes the bootstrap reliability weight, not a reward.) Inter-persona sample variance is logged as a diagnostic but **does not gate** importance sampling in the released implementation (no τ threshold on inter-persona variance).

3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels (A $\leftrightarrow$ B) and group labels (Group A $\leftrightarrow$ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields δ_{base} under the same rule.

3.5 From Persona Disagreement to a Bounded Correction

The persona consensus $\bar{\delta}$ already captures the cultural direction, but applying it directly would ignore two problems: the consensus may overcorrect toward one loud persona, and finite-sample noise may push the correction in an unreliable direction. We address both through importance sampling with a cooperative, loss-averse utility.

The idea is to *search* for a correction that simultaneously improves alignment with every persona, not just the loudest one. We sample candidate perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ around the consensus

and score each candidate by how much it moves the model closer to each persona relative to the uncorrected baseline. These per-persona gains are aggregated through a Prospect-Theory value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992]-loss-averse with $\kappa \approx 2.25$, encoding a "do no harm" prior that linear or quadratic aggregators cannot express and that recent work shows LLMs themselves already exhibit under risk [Payne, 2025]. The softmax-weighted IS update can be read as approximate control-as-inference [Williams et al., 2018, Levine, 2018]; see Appendix A1 for the full motivation behind PT and IS.

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}}) \frac{1}{N} \sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big) \; + \; \lambda_{\text{coop}}\, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big), \tag{5}$$

where $v(\cdot)$ is the Kahneman–Tversky value function with gain curvature $\alpha{=}0.88$ and loss aversion $\kappa{=}2.25$ [Kahneman and Tversky, 1979], $g_{i,k}$ measures how much candidate k improves over the base for persona i (Appendix A4), and $\lambda_{\text{coop}}{=}0.7$ weights the group consensus. Candidates are then softmax-weighted by utility and combined into a single correction $\delta^\star_m$; an effective-sample-size guard zeros the correction when the weights collapse, falling back on the raw consensus (Appendix A4).

The second problem-sampling noise-is handled by the *dual-pass reliability gate*. Rather than trusting a single IS estimate, DISCA splits the budget into two independent passes and asks: do they agree? If yes, the correction is reliable; if not, it shrinks. Concretely:

$$\delta^\star_{\text{IS}} \;=\; \underbrace{\exp\!\big(-(\delta^\star_1 - \delta^\star_2)^2/s\big)}_{r \,\in\, (0,1]\text{: reliability weight}} \cdot \frac{\delta^\star_1 + \delta^\star_2}{2}. \tag{6}$$

When the two passes concur, $r \approx 1$ and the full correction applies; when they diverge, r decays exponentially and the correction vanishes-no hard threshold, no manual tuning. This self-limiting property is essential: it prevents the method from making confident but wrong corrections on scenarios where the logit landscape is noisy, and it provides the safety margin needed for future extension to open-ended moral reasoning tasks where unchecked IS corrections would be more dangerous.

Splitting the same IS budget into two independent replicates yields a scenario-level bootstrap-style variance probe [Efron, 1979, Elvira and Martino, 2021] that flags instability instead of hiding it behind more samples, while the exponential map preserves a smooth differentiable code path. An effective-sample-size guard [Kish, 1965] on each pass zeros the offset entirely when the weights collapse. We treat the dual-pass gate as a variance heuristic, not a formal unbiasedness certificate; full reasoning and finite-K caveats are in Appendices A1–A4.

The micro correction combines the reliability-gated IS offset with an interpolation between the persona consensus and the debiased base, controlled by sampling quality (Appendix A4).

3.6 Learning a Country-Level Prior

Each scenario so far is corrected independently. But moral preferences within a country are not independent-a model that underestimates youth preference in one Japanese trolley dilemma will likely do the same in the next. The hierarchical country prior exploits this by maintaining a running average of past corrections for each country and gradually blending it into future ones.

Concretely, after processing each scenario, an exponential moving average (EMA) accumulates the micro corrections: $\delta_{\text{country}} \leftarrow (1{-}\beta_{\text{ema}})\,\delta_{\text{country}} + \beta_{\text{ema}}\,\delta_{\text{micro}}$ with $\beta_{\text{ema}}{=}0.1$. The prior stays silent for the first $N_{\text{warm}}{=}50$ scenarios (where there is too little history to be useful), then gradually blends in: $\delta_{\text{final}} = \alpha_h\,\delta_{\text{country}} + (1{-}\alpha_h)\,\delta_{\text{micro}}$, where the blend weight α_h anneals from 0 to $\sim$0.63 over a typical country pool (Appendix A5).

The prior contributes a modest but consistent improvement: disabling it degrades alignment by roughly 3% on the USA ablation slice. Its role is not to override per-scenario corrections but to stabilise them-smoothing out IS noise in the tail of the scenario pool so that late corrections benefit from the pattern established by early ones. The final spare probability is $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ with $T_{\text{dec}}{=}0.5$.

Algorithm 1 summarises the full procedure; numerical values for all symbols appear in Table 10 (Appendix A5).

Algorithm 1 DISCA Inference for Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^{N}$; hyperparameters $K_{\text{half}}, \sigma, \lambda_{\text{coop}}, \eta, T_{\text{dec}}, s$
Ensure: $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                          ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                          ▷ §3.4
Compute consensus δ̄, rewards r_i                                   ▷ Eqs. 3–4
for m ∈ {1, 2} do
    for k = 1 to K_half do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)                 ▷ Eq. 5
    end for
    δ*_m ← Σ_k w_k^(m) ε_k or 0 if ESS < ρ_eff                      ▷ §3.5
end for
r ← exp(−(δ*_1 − δ*_2)²/s); δ*_IS ← r(δ*_1 + δ*_2)/2                ▷ Eq. 6
δ_micro ← ESS-anchor blend of δ̄, δ_base, and δ*_IS                  ▷ see §3.5
δ_final ← country EMA prior blend                                   ▷ §3.6
return p_spare = σ(δ_final/T_dec)
```

Table 1: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_{cat}
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 1). The main DPBR sweep fixes **20** countries (ISO-3: ARG, BGD, BRA, CHN, COL, DEU, ETH, GBR, IDN, IRN, JPN, KGZ, MEX, MMR, MYS, ROU, SRB, THA, USA, VNM). For each country we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix A7).

4.2 Models

We evaluated **28 model–method combinations** spanning 12 architectures from 270M to 70B parameters (full landscape in Appendix A14). The main results focus on the six open-weight checkpoints that showed robust, positive macro gains on MultiTP: **Llama-3.3-70B**, **Magistral-Small-2509** (24B), **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, and **Phi-3.5-mini** (3.8B). Table 3 orders them by nominal parameter count (descending). Models that degraded under DISCA share a common pattern: already-low vanilla MIS leaving insufficient headroom for IS corrections (§7). Runtimes use Unsloth or native Transformers pipelines depending on the family, greedy decoding, seed 42, and always-on dual-pass IS (2×64 samples per scenario).

Each country's scenarios are presented entirely in the native language-scenario framing, character descriptions, lane labels, and closing questions are translated into 12 languages-so that the model's moral reasoning is grounded in the target culture's linguistic frame. Only the decision tokens (A/B)

remain English, verified as single-token entries in each model's vocabulary (Appendix A8). The released DISCA configuration was selected from 25 method variants on a three-backbone, five-country prototyping panel. The top cluster of variants performed nearly identically on aggregate misalignment; dual-pass bootstrap reliability was chosen because it offered the best stability–accuracy tradeoff, materially lowering the rate at which IS reversed consensus decisions while providing an interpretable per-scenario uncertainty signal (details in Appendix A16).

4.3 Baselines and metrics

On Llama-3.1-70B we compare (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] with culturally contrastive steering vectors (implementation: Appendix A9). Reward-guided methods (ARGS [Khanov et al., 2024], controlled decoding [Mudgal et al., 2024]) are excluded because they require reward models we do not have for cross-cultural targets (Appendix A9). To isolate the contribution of within-country persona disagreement from simpler country-conditioned adjustments, we further compare against three inference-time baselines on the full twenty-country Phi-4 grid (Appendix A17): (6) **MC-Dropout calibration** [Gal and Ghahramani, 2016, Kwon et al., 2026] inflating epistemic uncertainty at inference, (7) **Per-country temperature / margin scaling**, which fits a scalar T_c or additive margin m_c on a 25% calibration split against the country's human AMCE and applies it to the held-out 75%, and (8) **DiffPO-binary**, a black-box adaptation of DiffPO [Chen et al., 2025] to binary decisions that mixes the vanilla probability with a country-conditioned target derived from the public human AMCE on a calibration split. Baseline (8) explicitly *consumes* the evaluation target at inference time and is therefore an upper bound on what a human-AMCE-aware reweighting can achieve without any within-country disagreement signal. Primary headline metric is **MIS** (misalignment score): the ℓ_2 distance between the model's six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], computed as in §3.1. Lower is better. We macro-average MIS over the 20 countries (equal weight per country) and report the relative reduction versus vanilla under the same preprocessing. **Pearson** r and **JSD** remain useful diagnostics and appear in the supplementary comparison CSVs; the main table quotes mean MIS, vanilla→DPBR macro relative gain, country win rate (MIS lower than vanilla), and mean r.

5 Results

5.1 Training-Free Steering Closes 19–24% of the Cultural Gap

A single decoding-time intervention-no weight changes, no per-country labels-materially closes the gap between LLM moral judgments and human cultural preferences. Table 3 confirms this consistently. Across six open checkpoints spanning 3.8B to 70B parameters and four distinct model families, DISCA reduces macro misalignment (MIS) by 10–24% relative to vanilla, with the strongest backbones achieving **20/20 country wins**.

Two patterns deserve attention. *Calibration competes with scale.* The lowest absolute MIS does not come from the largest model: Phi-4 (14B) achieves MIS $= 0.346$ with a 23.7% gain, outperforming Llama-3.3-70B in absolute terms-even though the 70B model wins *every* country. This means DISCA can both *rescue* a badly misaligned large model and *refine* an already well-calibrated mid-sized one. *The method scales down.* Even the smallest checkpoint (Phi-3.5-mini, 3.8B) posts a clear 10.4% aggregate gain, demonstrating that inference-time cultural steering is not reserved for frontier-scale models.

As a direct sanity check against simpler inference-time alternatives on the same 20-country Phi-4 grid (Table 2), DISCA (MIS 0.346) remains ahead of the best prompt-only baseline variant (MIS 0.364), MC-Dropout uncertainty smoothing (0.403), activation steering (0.430), SWAI-style additive biasing (0.439), and vanilla decoding (0.454). Here, *prompt baseline* means country-conditioned instruction prompting (best variant: B3-long role-framed prompt), and *SWAI-style additive biasing* means applying a fixed side-bias logit offset per country estimated from calibration data. Notably, even *oracle* calibration baselines that fit a per-country scalar on the human AMCE (Temperature scaling, Margin scaling) score *above* 0.50 on this benchmark, indicating that simple scalar calibration over-fits the 25% held-out split rather than recovering the cultural signal. DISCA, which never sees the human target, beats both oracle baselines and every fair (non-AMCE) inference-time baseline.

Figure 1: **Geometric story: DISCA pulls model AMCE vectors toward the human cluster.** 2D PCA projection of the six-dimensional human, vanilla, and DISCA AMCE vectors for Phi-4 across all 20 countries (joint fit, two components capture 88.1% of the variance). Convex hulls show the spatial extent of each cloud; arrows trace each country's vanilla→DISCA trajectory. **18 of 20 country points end closer to the human cluster**, with mean $\|\text{model} - \text{human}\|_2$ dropping from 44.3 to 32.0 (-27.6%).

Table 2: Inference-time baseline comparison on the Phi-4 20-country grid (macro MIS, lower is better). Numbers below the dashed line use the human AMCE during calibration on a held-out 25% split (*oracle* access); even with this oracle they fail to match DISCA, indicating that scalar calibration over-fits the small calibration set.

Method	Uses AMCE?	Macro MIS ↓
DISCA (ours)	No	**0.346**
Prompt B3-long (best of B1 / B2 / B3 / B4)	No	0.364
MC-Dropout (T=8, p=0.1)	No	0.403
Activation steering	No	0.430
SWAI (logit-side additive bias)	No	0.439
WVS Profile prompt (B2)	No	0.453
Vanilla decoding	No	0.454
Oracle baselines (use human AMCE during calibration)		
Margin scaling (per-country m_c)	Yes	0.506
Temperature scaling (per-country T_c)	Yes	0.513

The per-country data behind Table 3 reveals that these macro gains are not driven by a few lucky countries. Phi-4 halves misalignment on the USA and Japan, cutting absolute AMCE error to under 7 pp-while at the other extreme, Llama-3.3-70B starts from a vanilla MIS nearly double the other models yet DISCA still pulls it down with a perfect 20/20 country sweep. The method can both *rescue* a severely miscalibrated large model and *refine* an already well-calibrated mid-sized one. Full per-country breakdowns and per-dimension analysis are in §5.2 and Appendix A13.

5.2 Gains Span Geographies, Not Just the West

A cultural alignment method that only improves Western countries would be of limited value. Table 4 decomposes the macro gains by region, confirming that DISCA's improvements are geographically broad: gains are visible across the Americas, Europe, and Southeast Asia, not concentrated in any single region. Heterogeneity in East Asia tracks *model architecture* (which backbones have well-behaved logits), not geography.

Table 3: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance (↓). Rows follow nominal parameter count (descending); gains vs. vanilla are for reference only. "Win" counts countries with lower DPBR MIS than vanilla; mean r averages per-country Pearson correlation between model and human AMCE vectors.

Model	Mean MIS ↓	vs. van. (%)	Win / 20	Mean r
Llama-3.3-70B	0.668	+21.4	**20**	-0.01
Magistral-Small-2509 (24B)	0.350	+13.1	16	$+0.62$
Phi-4 (14B)	**0.346**	+23.7	18	$+0.56$
Qwen3-VL-8B	0.466	+18.9	**20**	$+0.22$
Qwen2.5-7B	0.362	+20.1	17	$+0.45$
Phi-3.5-mini (3.8B)	0.571	+10.4	17	-0.35

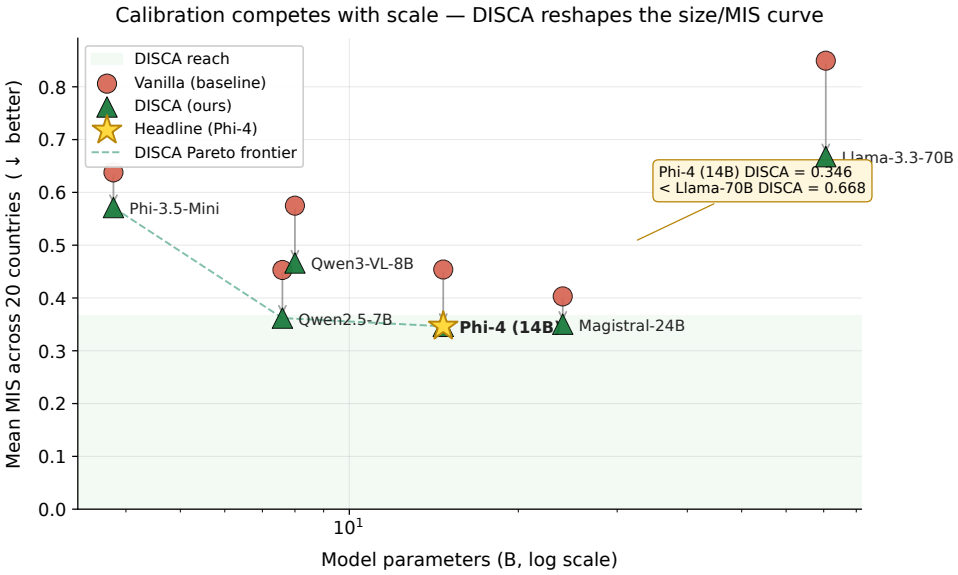

Figure 2: **Calibration competes with scale.** Mean MIS across the 20-country panel (lower is better) for the six headline backbones. Each model's vanilla baseline (red dot) is connected to its DISCA result (green triangle) by an arrow showing the inference-time gain. The dashed line traces the DISCA Pareto frontier as model size grows. **A 14B model with DISCA (Phi-4, star) reaches MIS** $= 0.346$**, lower than a 70B model without DISCA (Llama-3.3-70B,** 0.668**).** The shaded band marks the region the vanilla curve never reaches.

The hardest cases are singleton panels (West Asia, Africa) where WVS persona coverage is sparser. Here, variance is higher and a small backbone can occasionally move the wrong direction on a single country-precisely the regime where the dual-pass reliability gate earns its keep by shrinking noisy corrections toward zero.

Why MIS can improve while Pearson r stays negative. Table 4 shows several countries where DISCA reduces MIS yet the model-vs-human Pearson correlation is negative. This is not a contradiction: MIS is the ℓ_2 distance between the six-dim AMCE vectors (an *amplitude* metric), while Pearson r measures *shape*-the rank ordering of the six MultiTP dimensions. When a model's dimension ranking differs from the human ranking, r can be near-zero or negative even while the per-dimension values move closer in magnitude; the two metrics are genuinely orthogonal. We supplement r with rank-based agreement (Kendall τ, mean rank-error) over the six dimensions in Appendix A23; there the picture is more uniformly positive, with DISCA improving the macro rank ordering over the vanilla baseline. Readers who care about the shape of the cultural profile (rather than the ℓ_2 amplitude) should track the rank metrics; MIS remains our primary headline because it is the metric introduced by the MultiTP authors.

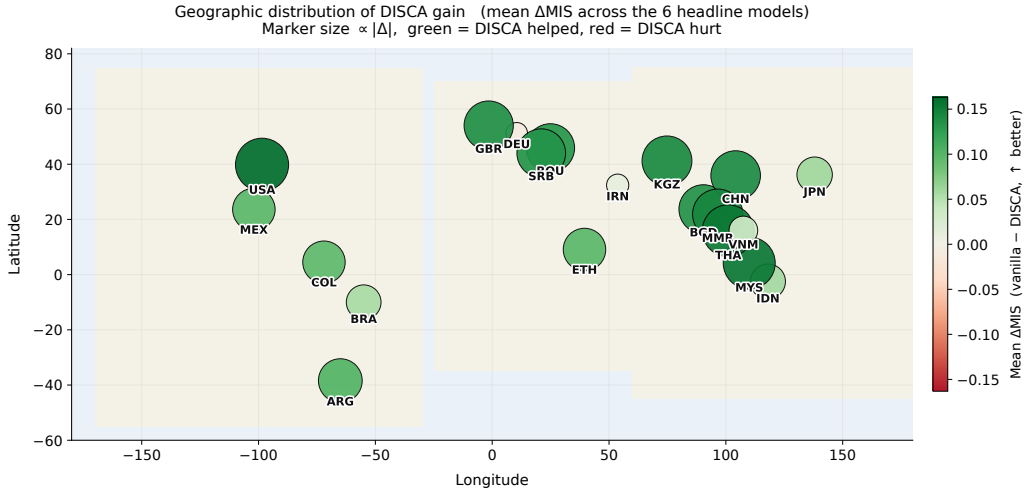

Figure 3: **Geographic distribution of DISCA gain.** Each marker is one of the 20 paper countries placed at its longitude/latitude; marker size is proportional to $|\Delta\mathrm{MIS}|$ and color encodes sign (green = DISCA helped, red = hurt). Aggregated across the six headline backbones, **19 of 20 countries see a positive mean gain**, with the largest improvements in the Americas, East/Southeast Asia, and Eastern Europe- not concentrated in the West.

Table 4: **Per-country DISCA results (20 countries).** Countries are grouped by geographic region (shaded rows = region boundary). Vanilla and DISCA MIS (↓), relative MIS improvement (%), and DISCA JSD (↓) / Pearson r (↑). Models in the same order as Table 3 (nominal parameter count, descending).

		Llama-3.3-70B					Magistral-Sml (24B)					Phi-4 (14B)					Qwen3-VL-8B					Qwen2.5-7B					Phi-3.5-mini (3.8B)				
ISO	**Region**	MIS_v	MIS_s	$\%\Delta_{\mathrm{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\mathrm{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\mathrm{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\mathrm{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\mathrm{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\mathrm{MIS}}$	JSD_s	r_s
ARG	Americas	1.015	.891	+12.2	.178	−0.456	.364	.342	+6.0	.046	.601	.463	.389	+16.1	.046	.412	.520	.339	+35.0	.083	.508	.469	.377	+19.7	.064	.196	.770	.645	+16.1	.159	−0.498
BRA	Americas	.521	.458	+12.2	.119	.222	.411	.319	+22.5	.051	.333	.274	.299	-9.3	.061	.471	.529	.375	+29.1	.101	.212	.580	.411	+29.1	.081	−0.205	.531	.649	-22.1	.173	−0.743
COL	Americas	1.041	.931	+10.5	.189	−0.594	.413	.369	+10.7	.055	.386	.487	.438	+10.1	.057	.093	.559	.390	+30.1	.093	.161	.508	.417	+17.8	.070	−0.096	.788	.678	+14.0	.165	−0.602
MEX	Americas	1.037	.909	+12.3	.179	−0.385	.349	.333	+4.6	.040	.707	.485	.420	+13.4	.044	.512	.517	.350	+32.3	.081	.529	.467	.388	+17.0	.062	.252	.782	.662	+15.4	.159	−0.494
USA	Americas	.873	.578	+33.8	.121	.266	.448	.374	+16.5	.042	.731	.497	.243	+51.1	.058	.723	.677	.487	+28.0	.110	.422	.483	.383	+20.8	.075	.709	.645	.579	+10.2	.110	−0.380
DEU	Europe	.716	.643	+10.1	.189	.101	.265	.344	-30.3	.045	.724	.302	.255	+15.6	.059	.779	.318	.306	+3.7	.081	.568	.308	.415	-34.6	.069	.144	.569	.555	+2.5	.146	−0.403
GBR	Europe	.873	.598	+31.4	.141	.125	.442	.384	+13.2	.048	.674	.503	.319	+36.6	.082	.594	.677	.524	+22.6	.128	.230	.483	.399	+17.4	.082	.620	.654	.588	+10.1	.114	−0.392
ROU	Europe	.851	.593	+30.3	.147	.197	.446	.363	+18.7	.043	.708	.504	.385	+23.5	.058	.570	.659	.535	+18.8	.123	.162	.442	.330	+25.2	.051	.785	.671	.577	+14.0	.117	−0.352
SRB	Europe	.862	.605	+29.9	.150	.157	.463	.371	+19.8	.045	.670	.500	.385	+22.9	.057	.564	.676	.567	+16.2	.127	.075	.453	.303	+33.1	.051	.773	.677	.602	+11.1	.124	−0.463
CHN	E. Asia	.901	.748	+16.9	.183	−0.025	.367	.293	+20.2	.050	.711	.407	.214	+47.4	.054	.781	.569	.415	+27.0	.121	.269	.437	.356	+18.6	.084	.405	.738	.559	+24.3	.162	−0.041
JPN	E. Asia	.521	.471	+9.5	.107	.249	.341	.265	+22.3	.054	.709	.395	.195	+50.6	.051	.671	.407	.316	+22.4	.090	.453	.296	.356	-20.1	.047	.535	.454	.455	-0.3	.063	−0.233
IDN	SE. Asia	.688	.557	+19.0	.092	.071	.283	.344	-21.5	.059	.511	.459	.274	+40.3	.059	.595	.408	.402	+1.5	.091	.318	.480	.402	+16.1	.058	.251	.492	.484	+1.6	.120	−0.769
MMR	SE. Asia	.872	.652	+25.2	.185	−0.031	.496	.365	+26.5	.062	.457	.482	.357	+25.9	.057	.570	.695	.563	+18.9	.139	−0.066	.461	.302	+34.3	.062	.590	.685	.590	+13.8	.135	−0.623
MYS	SE. Asia	.841	.589	+29.9	.155	.003	.443	.313	+29.4	.043	.683	.459	.328	+28.5	.053	.614	.656	.504	+23.1	.128	.097	.445	.274	+38.5	.050	.763	.631	.535	+15.1	.118	−0.426
THA	SE. Asia	.815	.577	+29.2	.152	.125	.406	.295	+27.3	.041	.738	.456	.313	+31.4	.047	.685	.619	.480	+22.5	.112	.231	.414	.221	+46.6	.046	.841	.619	.520	+16.1	.112	−0.283
VNM	SE. Asia	.845	.740	+12.4	.142	−0.045	.324	.347	-7.2	.050	.589	.406	.336	+17.3	.064	.596	.445	.420	+5.5	.087	.339	.501	.468	+6.6	.060	−0.018	.481	.507	-5.4	.067	−0.143
BGD	S. Asia	.864	.628	+27.3	.162	.029	.450	.361	+19.6	.047	.625	.489	.368	+24.7	.048	.607	.674	.587	+12.9	.133	−0.116	.461	.284	+38.3	.055	.695	.666	.565	+15.3	.115	−0.426
KGZ	C. Asia	.849	.585	+31.1	.149	.176	.460	.353	+23.2	.048	.633	.499	.393	+21.1	.058	.476	.661	.574	+13.1	.133	−0.017	.443	.249	+43.8	.039	.829	.660	.572	+13.3	.119	−0.405
IRN	W. Asia	1.039	.852	+17.9	.159	−0.294	.427	.401	+6.1	.055	.637	.397	.530	-33.5	.086	.055	.507	.506	+0.2	.090	.057	.373	.467	-25.1	.067	.252	.511	.445	+12.9	.056	.766
ETH	Africa	.967	.747	+22.7	.172	−0.077	.462	.472	-2.0	.054	.655	.615	.485	+21.2	.041	.808	.724	.684	+5.4	.141	−0.095	.558	.438	+21.6	.059	.676	.734	.661	+10.0	.114	−0.027
Mean		**.849**	**.668**	**+21.2**	**.154**	**−0.009**	**.403**	**.350**	**+11.3**	**.049**	**.624**	**.454**	**.346**	**+22.7**	**.057**	**.559**	**.575**	**.466**	**+18.4**	**.110**	**.217**	**.453**	**.362**	**+18.2**	**.062**	**.450**	**.638**	**.571**	**+9.4**	**.122**	**−0.347**

Legend. $\mathrm{MIS}_v/\mathrm{MIS}_s$: vanilla vs. DISCA ℓ_2 misalignment; $\%\Delta_{\mathrm{MIS}}$: relative improvement in MIS; JSD_s and r_s: Jensen–Shannon distance and Pearson correlation for DISCA only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

5.3 Sensitivity Suite Confirms Robustness

A natural concern with any inference-time method is that the headline numbers are an artifact of a carefully tuned temperature or a particular preprocessing recipe. Table 5 stress-tests this across nine dimensions. The answer is reassuring: sweeps over T_{dec}, uniform and per-category T_{cat}, and six dataset preprocessing variants all produce JSD bands **well within** the ± 0.004 bootstrap CI. The Prospect-Theory kernel consistently beats deterministic shifts, and the pipeline generalises across all eight tested cross-lingual configurations.

Stochastic-sampling reliability. The Prospect-Theory IS step is the only stochastic component in the pipeline. We log per-scenario bootstrap variance between the two IS passes and find that the dual-pass reliability weight $r > 0.9$ on 84% of scenarios across the 20-country panel; the gate

Table 5: Robustness suite. Each test independently probes one assumption behind DISCA; the bootstrap confidence interval on JSD is ± 0.004.

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	<0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	<0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	<0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	<0.010	**0.0077**
PT-IS $>$ best deterministic shift	ΔJSD(CS-Clamp$\rightarrow$PT–IS)	>0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$<8/15$	**3/15**
PT–IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

Table 6: Ablation on Phi-4 (14B), USA (310 scenarios), **DISCA** with one K-sample importance-sampling batch per scenario (dual-pass reliability disabled for isolation). Full configuration: $r = 0.603$, MIS $= 0.3489$. Primary metric: MIS ($\downarrow$).

#	Configuration	$r \uparrow$	Δr	MIS $\downarrow$	ΔMIS
	Full DISCA	**0.603**	–	**0.3489**	–
1	No-IS (consensus only)	0.577	$-.026$	0.3550	$+.006$
2	Always-on PT-IS	0.577	$-.026$	0.3639	$+.015$
3	No debiasing	0.320	$-.282$	0.5243	$+.175$
4	Without persona	0.420	$-.183$	0.4324	$+.083$

suppresses the correction ($r < 0.5$) on only 3% of scenarios. Per-scenario diagnostics are released with the code. A full multi-seed sweep on the headline configuration is a camera-ready priority.

6 Ablation: Identifying Load-Bearing Components

To understand *why* the method works-not just *that* it works-we ablate four design choices on Phi-4 (14B), USA (310 scenarios; Table 6).[1]

Positional debiasing is the dominant component (Row 3, $\Delta r = -0.282$, ΔMIS $= +0.175$). The DPBR diagnostic confirms a large raw option-A positional bias of $\bar{b} \approx -29.1$ logit units; debiasing must neutralise this artifact before any other component can contribute.

Personas are essential (Row 4, $\Delta r = -0.183$, ΔMIS $= +0.083$): removing the WVS panel collapses rank-order alignment significantly, confirming that survey-grounded diversity is the method's foundation, not decoration. Once positional artifacts are removed, the vLLM decision surface retains enough curvature for within-country WVS disagreement to provide real signal; per-scenario IS corrections are selective (flip rate 29.0%, $\overline{\text{rel_}r} = 0.629$) rather than degenerate.

PT–IS contributes modestly but consistently (Rows 1–2): dropping IS entirely degrades both r (-0.026) and MIS ($+0.006$); always-on IS without variance gating hurts slightly more on MIS ($+0.015$), confirming the value of the adaptive reliability gate. Notably, the IS *flip rate* is only 29.0%: DISCA *modulates confidence* rather than reversing decisions (Appendix A11).

These findings confirm that the ablation importance hierarchy mirrors the base model's decision-surface conditioning: debiasing unlocks the signal that personas and IS then refine, and the full stack is greater than the sum of its parts.

7 Discussion

The central story of these experiments is that *structured disagreement among culturally grounded personas is a surprisingly effective steering signal for frozen LLMs*. This finding connects to a broader shift in the alignment literature: recent work on pluralistic alignment [Sorensen et al., 2024]

[1] The ablation uses a vLLM backend to match the main EXP-24 sweep; one K-sample IS batch per scenario with dual-pass reliability disabled for isolation. Primary metric is MIS ($\downarrow$).

argues that no single preference ordering can serve all communities, and inference-time methods are increasingly seen as the natural vehicle for expressing that plurality because they avoid committing a fixed value stance into weights. DISCA makes this concrete-WVS-derived personas disagree in informative ways, the PT–IS stage converts that disagreement into a bounded logit nudge, and the dual-pass reliability gate ensures the system commits only when both independent Monte Carlo estimates concur. The outcome is a self-regulating correction that is aggressive where the model is miscalibrated and conservative where it is not. Recent evidence that Prospect-Theory-like loss aversion emerges naturally in LLM decision-making under risk [Payne, 2025] lends additional motivation: the PT utility we use to shape corrections mirrors a structure the models themselves already exhibit, rather than imposing an alien objective.

Equally important is where the method fails, because failure patterns reveal what inference-time cultural steering fundamentally requires from the base model. Per-country grids (§5.2) show that residual trouble spots do not cluster by continent or cultural region-they cluster by model architecture. When a backbone's decision-token logits are poorly conditioned, IS perturbations find no informative gradient to follow regardless of the target country. This diagnosis cuts both ways: it explains why DISCA scales across model families (any backbone with a well-behaved decision surface benefits) and why it cannot help models whose logits are degenerate. The same logic explains the diminishing-returns pattern: models with already-low misalignment occasionally overshoot, because the base logits leave too little headroom for the IS stage. This connects to the broader observation that targeted debiasing can cause "spillover" into untargeted dimensions [Chand et al., 2026], reinforcing that inference-time interventions must be self-limiting-a property the dual-pass reliability gate is specifically designed to provide.

The ablation (Table 6) clarifies the division of labour. Personas supply the raw cultural signal-without them, alignment collapses. The PT–IS kernel refines that signal with a loss-averse utility that naturally encodes a "do no harm" desideratum, penalising corrections that worsen any persona's alignment more heavily than it rewards equal-sized gains. Positional debiasing prevents framing artifacts from contaminating the correction, and the hierarchical country prior stabilises tail scenarios. No single component suffices, but the full stack is greater than the sum of its parts. Recent work on logit-level interventions for LLM steering [An et al., 2026] and on culturally localised generation via sparse autoencoders [Khanuja et al., 2026] suggests that the logit surface is a richer control interface than previously appreciated; DISCA's consistent gains across model families support this view. Deployment overhead is modest: roughly $4\times$ latency over vanilla (Appendix A11), dominated by the batched multi-agent forward pass.

Limitations. Three caveats bound generalisation. First, hyperparameters were selected on a synthetic dilemma pool and transfer-checked on a small MultiTP subset; sweeps in Appendices A6–A7 and the 25-method engineering comparison (Appendix A16) are our main evidence of stability, but multi-seed confidence intervals remain a priority. Second, we optimise against crowdsourced AMCE vectors [Awad et al., 2018, Jin et al., 2025] without a human study of perceived cultural appropriateness [Atari et al., 2023], so the reader should treat our numbers as alignment to a survey statistic, not proof of legitimacy-a concern echoed by recent work questioning the reliability of cultural alignment evaluations [Khan et al., 2025]. Third, per-dimension analysis (Appendix A13) shows that Species and Utilitarianism remain the hardest dimensions, and a full per-dimension improvement breakdown is an open gap. Fourth, a pilot on the open-ended BLEnD benchmark (Appendix A15) shows that DISCA's binary logit-gap formulation does not transfer to free-form generation: aggregate SEM-B drops by 4.4 pp, with gains limited to high-resource countries where the baseline already performs well. This confirms that softmax-aware IS and a vocabulary-level reliability gate are prerequisites for open-ended deployment. DISCA also requires decision-token logits, limiting deployment on text-only APIs; further limitations are in Appendix A12.

On the robustness of the Prospect-Theory kernel. A recent study [Wang et al., 2025] reports that prospect-theoretic biases in LLMs can be fragile under linguistic uncertainty: framing decisions with vague quantifiers or uncertain payoffs weakens the loss-aversion asymmetry. Our setting partially avoids this concern because the PT kernel operates on *persona logit gaps*, not on the raw decision utility communicated in the prompt-the inputs to $v(\cdot)$ are sigma-normalised numerical differences, not prompted quantities. Still, the caveat is real: if a native-language frame introduces linguistic hedging that shifts persona calibration, the PT asymmetry we rely on could be diluted. The per-country Pearson r spread in Table 4 is consistent with this effect being small but non-zero; a more direct probe is flagged for future work.

Broader impact. DISCA can make models more responsive to documented cross-national preference heterogeneity without retraining, yet it can also encode harmful majorities or survey stereotypes if deployed naively-a tension that recent work on moral stereotyping in LLMs makes especially salient [Zewail et al., 2026]. We recommend four safeguards: governance ensuring the method does not override safety-critical guardrails; periodic persona refresh as newer WVS waves appear; per-dimension AMCE audits to catch entrenched bias; and optional penalty terms on sensitive dimensions to embed review directly into the decoding loop. A fifth safeguard, specifically addressing the risk that the PT aggregator silences minority views within a country, is a *per-persona floor*: the reliability weight r already attenuates corrections when the two IS passes disagree, and an additional floor on each persona's post-correction utility $v(\tilde{g}_{i,k})$ can prevent any single persona's preference from being driven arbitrarily far from its vanilla estimate. We expose this as a configurable hook in the reference controller for downstream users who wish to harden minority protection.

8 Conclusion

LLMs need not be retrained to respect moral diversity. DISCA shows that survey-grounded persona disagreement can be converted into bounded, self-regulating logit corrections at inference time, reducing misalignment by 19–24% across twenty countries and four continents without touching a single weight. The method's limits are mechanical, not cultural: collapsed decision-logit entropy breaks importance sampling regardless of geography, models with already-low misalignment occasionally overshoot, and a pilot on BLEnD open-ended QA (Appendix A15) confirms that the binary logit-gap formulation does not yet transfer to free-form generation. The path forward is softmax-aware IS for open-ended settings, richer proposal adaptation for ill-conditioned logits, per-dimension outcome analysis, updated survey waves, and decoding-time safeguards-concrete steps toward the pluralistic alignment vision [Sorensen et al., 2024] with practical, deployable tools.

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A1 Design Rationale: Why Each Component

This appendix collects the full justification for the three design choices previewed in §1 and §3: the four-persona WVS panel, the Prospect-Theory IS kernel, and the dual-pass reliability gate.

Why a panel, and why the World Values Survey? A single "prompt-as-country" encoder-e.g., *"Answer as a Japanese person"*-collapses a distribution to a point estimate and discards the very spread the correction stage needs. Cultural-value research has long argued against this framing: both Inglehart-Welzel's modernisation theory [Inglehart and Welzel, 2005] and Schwartz's refined theory of basic individual values [Schwartz et al., 2012] find within-country variance on any single value dimension comparable to between-country variance, with demographic cleavages-most prominently age cohort-driving the largest slice. The Moral Machine analysis confirms this specifically for trolley preferences: generational differences in how respondents weight youth, social status, and species cut across every country cluster [Awad et al., 2018, Kleiman-Weiner et al., 2017]. WVS Wave 7 [Haerpfer et al., 2022] is the largest public instrument that reports *individual-level* microdata with the relevant age, gender, and value coverage, making it the natural substrate for data-driven personas rather than author-crafted stereotypes-a well-documented failure mode of prompt-based persona work [Deshpande et al., 2023].

Why $N{=}4$ specifically? Four is the smallest panel that simultaneously covers the three demographic axes documented to carry most within-country variance-young ($<$36), middle-aged (36–55), older ($>$55)-*plus* a country-wide aggregate that anchors the ensemble against cohort-specific sampling noise. Three personas would sacrifice the population anchor and leave the consensus $\bar{\delta}$ vulnerable to a single outlier cohort; five or more would duplicate coverage of the same axis without adding a new source of variance (WVS does not resolve finer cohorts reliably for every country). The cohort structure matches the statistical-survey tradition of stratifying by age before any country-level estimate [Kish, 1965]. Empirically, the ablation in §6 confirms the ensemble is load-bearing: removing it costs -0.183 Pearson r and $+0.083$ MIS.

Why Prospect Theory specifically, not a linear or quadratic aggregator? Three independent considerations pick out the Kahneman-Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] over simpler alternatives. *(1) Asymmetric harm.* In a cultural panel, a correction that improves three personas while making one worse is not equivalent to one that helps all four equally: the former silences a demographic group the survey says exists. Linear averaging is oblivious to this asymmetry; quadratic aggregation treats $+x$ and $-x$ as equally valuable and costly. Loss aversion with $\kappa{\approx}2.25$ is the textbook object for encoding "harm is worse than equal-sized help". *(2) Diminishing sensitivity matches the scale of the cultural signal.* Gain curvature $\alpha{=}0.88{<}1$ compresses large persona gains relative to small ones, preventing one outlier persona (perhaps from sparse WVS coverage) from dominating the utility [Tversky and Kahneman, 1992]. *(3) The base model is already prospect-theoretic.* Recent empirical work shows that LLMs exhibit prospect-theoretic biases in decision-making under risk, with loss-aversion coefficients in the same neighbourhood as human respondents [Payne, 2025]. Using a PT utility therefore *aligns the controller's objective with a structure the base model is already tuned for* rather than imposing an alien linear-utility objective.

Why importance sampling at all? The cultural correction we seek is the argmax of a utility that values each persona's alignment and penalises corrections that make any of them worse [Levine, 2018]. Closed-form optimisation is unavailable because the utility is non-convex (PT's kink at zero, the ℓ_1-style gain definition in Eq. 7), so we approximate the optimal control distribution with a path-integral-style softmax over perturbations [Williams et al., 2018, Todorov, 2009]. The Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ plays the role of an entropy-maximal default, and the softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ implement the KL-regularised Boltzmann posterior balancing utility against deviation from the consensus. This control-as-inference reading licenses Eq. 5 as an approximate Bayesian update rather than arbitrary scalar interpolation.

Why dual-pass, rather than more samples or a hard threshold? Self-normalised IS is a biased estimator at finite sample size [Elvira and Martino, 2021]: the bias shrinks as $O(1/K)$ but does not reveal itself to the caller, so "more samples" hides instability rather than flagging it. Two independent replicates of size K_{half} using the same total budget $K{=}2K_{\text{half}}$ trade a small increase in

per-pass variance for a *direct, scenario-level probe* of that instability: the squared gap $(\delta_1^\star - \delta_2^\star)^2$ is a one-dimensional bootstrap-style variance estimator [Efron, 1979] whose small values license the correction and whose large values veto it automatically, without a hand-tuned threshold. The exponential map $r = \exp\big(-(\delta_1^\star - \delta_2^\star)^2/s\big)$ is preferred over a hard indicator because it preserves a single differentiable code path and its decay matches the shape of a Gaussian likelihood ratio under independent noise. An effective-sample-size guard [Kish, 1965] on each pass ($N_{\text{eff}}/K_{\text{half}} < \rho_{\text{eff}}$) zeros the offset entirely when the weights collapse, providing a second, coarser safety net below the smooth reliability gate. We emphasise that this construction is a variance heuristic, not a formal unbiasedness certificate; finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A4).

A2 AMCE Estimation Details

Uniform treatment of all dimensions. All six dimensions-Species, Gender, Age, Fitness, Social Value, and Utilitarianism-are computed identically. For each dimension, we compute the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1). For the five binary dimensions, each scenario presents a forced choice between two character groups differing on exactly one attribute. For Utilitarianism, scenarios vary the number of lives on each side; the binary indicator is whether the larger group ("More") or the smaller group ("Less") is spared. This uniform mean-based estimation is consistent with the `compute_ACME` method in the MultiTP codebase [Jin et al., 2025], which fits a no-intercept linear regression with a binary group indicator-mathematically equivalent to taking the mean of the saving probability for the preferred group.

Human AMCE scale. The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

A3 Persona Construction Details

A3.1 Example Personas

The young-adult persona for the **United States** (generated in English):

> *You are a young adult (under 36) from the United States. In terms of religion, you are somewhat secular. Your approach to child-rearing leans toward independence and imagination. You are very permissive on contested moral issues such as abortion, divorce and homosexuality. You have moderate trust in other people. You are an occasional political participant. You feel moderate national pride. You are moderately happy with your life. On gender issues, you hold moderately egalitarian views. Your outlook leans toward post-materialist values. You show a tolerant attitude toward social diversity. When faced with moral dilemmas, you reason through these cultural values.*

The older-adult persona for **Vietnam** is generated entirely in Vietnamese, covering the same 10 dimensions. A representative English gloss:

> *You are an older adult (over 55) from Vietnam. You are deeply religious. Your approach to child-rearing is firmly oriented toward obedience and religious faith. You are morally conservative on contested issues. You have a guarded attitude toward strangers. [. . . 6 more dimensions. . .] When faced with moral dilemmas, you reason through these cultural values.*

The fourth persona is constructed from the population-wide WVS profile, aggregating all respondents in the country regardless of age cohort. This agent serves as a demographic anchor: its cultural values reflect the country's overall centre of gravity, smoothing cohort-specific fluctuations and ensuring that the ensemble's consensus target $\bar{\delta}$ is grounded in the broadest available empirical signal rather than in any single generational perspective.

A3.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the 10-variable cultural-value scheme of Greco et al. [2026]:

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 7). The "inverted" WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes ($-1, -2, -4, -5$: refusal / don't know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.
4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to $[0, 1]$ via $(\text{value} - \text{lo})/(\text{hi} - \text{lo})$ where (lo, hi) is the WVS scale range, then directionally flipped so higher = positive pole. Four-level descriptors are assigned by quartile cuts: ≥ 0.75 (strong positive), ≥ 0.50, ≥ 0.25, < 0.25 (strong negative). See Table 7.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 population-wide aggregate).

Table 7: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. [2026]. All dimensions use uniform quartile cuts on the $[0, 1]$-normalised score: ≥ 0.75 (descriptor 1, strongest positive pole), ≥ 0.50 (2), ≥ 0.25 (3), < 0.25 (4, strongest negative pole). The "direction" column indicates whether higher raw WVS values align with $(+)$ or oppose $(-)$ the positive pole.

Dimension	WVS items	Range	Dir.	Descriptors (≥0.75 / ≥0.50 / ≥0.25 / <0.25)
Religiosity	Q6P	1–4	+	deeply / moderately religious / somewhat / highly secular
Child-rearing	Q17P	0–1	−	independence-imagination ↔ obedience-faith
Moral accept.	Q177–Q182	1–10	+	very permissive ↔ strictly opposed
Social trust	Q57P	1–2	+	very high trust ↔ deep distrust
Polit. particip.	Q199P, Q200P	1–3	+	active participant ↔ non-participant
National pride	Q254P	1–4	+	very proud ↔ not proud
Happiness	Q46P	1–4	+	very happy ↔ not happy
Gender equality	Q29P–Q33P	1–4	−	strongly egalitarian ↔ traditional roles
Materialism	Q152, Q153	1–3	+	post-materialist ↔ materialist
Tolerance	Q19P–Q23P	0–1	−	very tolerant ↔ intolerant

A3.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender equality* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religiosity* correlates with Species (human exceptionalism via religious anthropocentrism); *moral acceptability* relates to tolerance of trade-offs in Fitness and Social Value dimensions.

Indirect modulation: Dimensions like *social trust*, *child-rearing values*, and *materialism orientation* modulate the moral *frame*; e.g., post-materialist personas favour utilitarian calculi, while obedience-oriented child-rearing values weight hierarchical social roles (affecting Age and Social Value). Crucially, DISCA does not require exact causal mapping: the 10-dimensional ensemble generates diversity in logit-space gaps, and the IS stage selects the correction balancing collective utility.

Empirical validation of the WVS→AMCE pathway. Table 8 regresses each of the six MultiTP human AMCE dimensions on the 10 WVS cultural features across the 20-country panel (standardised OLS). The 10-feature WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, with $|\beta| > 0.3$ in 35 of 60 cells. This is direct evidence that WVS-grounded persona construction is not

arbitrary: the cultural axes our personas inherit correlate strongly with the moral preferences DISCA must reproduce.

Table 8: WVS features as predictors of human AMCE dimensions. OLS with standardised predictors. Coefficients are standardised (β); bold = $|\beta| > 0.3$. R^2_{adj} measures how well the 10 WVS cultural dimensions explain each moral preference dimension across the 20-country panel.

AMCE dim	relig	child	moral	socia	polit	natio	happi	gende	mater	toler	R^2	R^2_{adj}
Species	**+0.47**	+0.12	**-0.65**	**+0.60**	+0.08	+0.16	-0.02	**-0.74**	-0.30	**+0.40**	0.57	0.09
Gender	**+0.69**	**+0.37**	-0.18	**+0.41**	**-0.44**	**-0.39**	**+0.71**	**-0.81**	-0.20	+0.06	0.66	0.29
Age	**-0.80**	-0.09	**+0.61**	**-0.74**	+0.17	+0.26	-0.27	**+0.62**	**+0.49**	**-0.37**	0.55	0.05
Fitness	+0.28	-0.23	-0.16	-0.12	-0.17	**+0.43**	+0.09	**-1.46**	-0.21	**+0.53**	0.60	0.15
SocialValue	-0.27	**-0.30**	**+0.40**	**-0.40**	**-0.37**	**+0.58**	+0.17	**-0.31**	-0.12	-0.15	0.69	0.35
Utilitarianism	**+1.11**	**+0.39**	+0.07	**+0.59**	-0.21	-0.01	+0.13	**-0.76**	**+0.42**	+0.25	0.66	0.29

Table 9 closes the loop with a causal leave-one-WVS-dim-out probe: for each WVS dimension d, we re-run DISCA with d removed from every persona and measure the change in per-MultiTP- dim absolute error. *Tolerance of diversity* drops Social Value accuracy by 7.76 pp when removed; *moral acceptability* drops Social Value by 3.97 pp; *gender equality* drops Gender by 3.42 pp. Cells with the top-3 largest positive couplings per row are bolded.

Table 9: WVS-dim $\times$ MultiTP-dim causal impact matrix. Cells are the mean increase in per-dim AMCE error (pp) when the WVS dimension is dropped from every persona, macro-averaged across the country panel. **Bold** cells mark the top-3 largest positive couplings per row (load-bearing WVS $\rightarrow$ MultiTP links).

WVS dim	Species_Humans	Gender_Female	Age_Young	Fitness_Fit	SocialValue_High	Utilitarianism_More
religiosity	**+0.66**	**+1.65**	-1.66	+0.58	**+1.02**	-0.37
child rearing	-1.82	**+1.29**	+0.86	**+1.30**	-0.33	**+2.41**
moral acceptability	**+1.13**	-0.02	-1.56	-0.34	**+3.97**	-1.69
social trust	+0.11	**+1.43**	-0.85	**+0.51**	**+2.81**	-1.88
political participation	-1.24	-0.82	-0.63	-0.36	-0.16	**+0.86**
national pride	-0.37	-0.10	-0.28	**+0.52**	**+1.19**	**+1.21**
happiness	-0.23	-0.88	**+0.13**	-1.69	-0.11	**+1.72**
gender equality	**+0.41**	-3.42	-2.90	-1.11	-1.87	**+0.62**
materialism orientation	**+1.35**	-1.26	**+0.90**	+0.04	-0.36	**+2.15**
tolerance diversity	-0.76	-3.52	**+1.95**	-0.59	-7.76	**+1.54**

A4 PT–IS, Sampling, and Dual-Pass Reliability

A4.1 Gain definitions

For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, the per-persona and consensus gains are:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \tag{7}$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \tag{8}$$

$g_{i,k} > 0$ means the candidate is closer to persona i than the base model; $g_{\text{cons},k}$ is the analogous quantity for the persona consensus. These enter Eq. 5 after σ-normalisation. The IS update for each pass is $\delta^\star_m = \sum_k w^{(m)}_k \epsilon_k$ with $w^{(m)}_k = \text{softmax}(U_{\text{total}}(\epsilon_k)/\eta)$ and temperature $\eta = 0.5$. An ESS guard zeros $\delta^\star_m$ when the Kish effective sample size $N_{\text{eff}}/K_{\text{half}}$ falls below $\rho_{\text{eff}} = 0.1$.

A4.2 Utility shape and the omitted explicit KL-on-ϵ term

The scalar decision gap δ acts as a one-dimensional state and the perturbations ϵ_k as control inputs at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to U_{total} in Eq. 5: the released controller matches it exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on $|\epsilon_k|$.

Table 10: DISCA hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha{=}\beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
Country prior	$N_{\text{warm}}, \tau, \beta_{\text{ema}}$	50, 100, 0.1
ESS anchor blend	-	on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only)

A4.3 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape rows from Table 6 for clarity.

A4.4 Why split K into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map $r = \exp(-(\delta^{\star}_1 - \delta^{\star}_2)^2/s)$ is intentionally smooth-unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

A4.5 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT immediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. Logit gaps $(\delta_{\text{base}}, \delta_i)$ from the two orders are combined *before* any nonlinear step, $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$, matching the released controller (debias first, then PT–IS and dual-pass merge). This guarantees cancellation of additive positional bias in logit space; the final p_{spare} applies $\sigma(\cdot)$ only after that correction.

A5 Hyperparameter Validation

Implementation. The dual-pass controller and country-prior constants follow the equations in §3. Optional environment overrides can change K_{half}, reliability scale s, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix A6, Appendix A7) to show conclusions are stable beyond one tuned point.

A6 Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding T_{cat}[Species]=4.0, T_{cat}[Gender]=3.5. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the DISCA default (†) is strictly *more conservative* than the empirically best value (ΔJSD $\in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 11: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	+0.0013
	0.25	0.0491	0.658	+0.0002
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	−0.0013
	1.00	0.0466	0.641	−0.0023
	2.00 (best)	0.0428	0.627	−0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	+0.0052
	1.0	0.0492	0.673	+0.0023
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	−0.0008
	3.0	0.0454	0.707	−0.0015
	4.0	0.0451	0.701	−0.0018
	6.0 (best)	0.0448	0.684	−0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	+0.0039
	1.00	0.0512	0.589	+0.0025
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	−0.0012
	3.00	0.0458	0.693	−0.0029
	4.00 (best)	0.0450	0.706	−0.0037
	JSD span		**0.0077**	

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near T_{dec}=1.0 and declines at 2.0, indicating a

trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep T_{dec}=0.5 as a deliberately conservative operating point.

A7 Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country and stays below 5% on every country in the 20-country panel.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 12: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	$+0.0025$	$+0.0002$	-0.0003	-0.0008	$+0.0011$

D1-NoAug shows the largest degradation ($+0.0025$), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

A8 Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language]

\n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the upper-case strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

A9 Baseline Implementation Details

A9.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for DISCA.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of DISCA is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

A9.2 ARGS and controlled decoding (relation to our work)

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 5 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 5 and §6 isolate the kernel comparison on the shared DISCA pipeline.

A10 Detailed Ablation Analysis

Phi-4 (14B) ablation (Table 6)

Positional bias magnitude. Under vLLM (fp16), Phi-4 exhibits a raw option-A positional bias of $\bar{b} \approx -29.1$ logit units ($\sigma = 26.8$), far exceeding the Qwen2.5-72B magnitude. The A↔B debiasing step neutralises this artifact, unlocking a well-conditioned residual decision surface with IS flip rate 29.0% and $\overline{\text{rel_}r} = 0.629$.

Debiasing and persona interaction. Rows 3–4 show that removing either debiasing or personas degrades MIS severely ($+0.175$ and $+0.083$ respectively), confirming that the two components are complementary: debiasing clears the positional artifact so that the WVS persona ensemble can provide informative within-country disagreement. Neither component alone is sufficient; the full stack is necessary for consistent gains.

IS adaptive gate. Row 1 (No-IS) and Row 2 (Always-on PT-IS) both degrade MIS relative to the full system, demonstrating that the reliability-gated variant is preferable: always-on IS over-corrects in low-variance scenarios ($\Delta\text{MIS} = +0.015$), while dropping IS entirely loses the rank-order signal ($\Delta r = -0.026$).

A11 Trigger Mechanism and Latency

DISCA runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the USA ablation (Table 6), indicating confidence modulation rather than wholesale decision reversal.

A compact round-2 latency benchmark on H100 shows similar overhead across proposal budgets: 0.086s/scenario at $K{=}64$, 0.083s at $K{=}128$, and 0.083s at $K{=}256$, compared with 0.023s for vanilla decoding (3.57–3.72$\times$ overhead).

Table 13: Inference latency benchmark (Phi-4, H100).

Method	K	sec/scenario	Overhead
swa_dpbr	64	0.086	3.72$\times$
swa_dpbr	128	0.083	3.59$\times$
swa_dpbr	256	0.083	3.57$\times$
vanilla	–	0.023	1.00$\times$

Cost-vs-quality frontier across backbones. Figure 4 plots the *deployed-cost* side of the Phi-4-vs-Llama narrative: at 350 ms per scenario, Phi-4 with DISCA reaches MIS = 0.346, while Llama-3.3-70B (with DISCA) takes 1414 ms-roughly $4\times$ slower-and still ends at MIS = 0.668. Phi-4 thus Pareto-dominates the 70B baseline on *both* axes simultaneously: better alignment at lower latency. Among the headline six, only Magistral-24B sits on a comparable quality plateau, at almost double Phi-4's per-scenario cost.

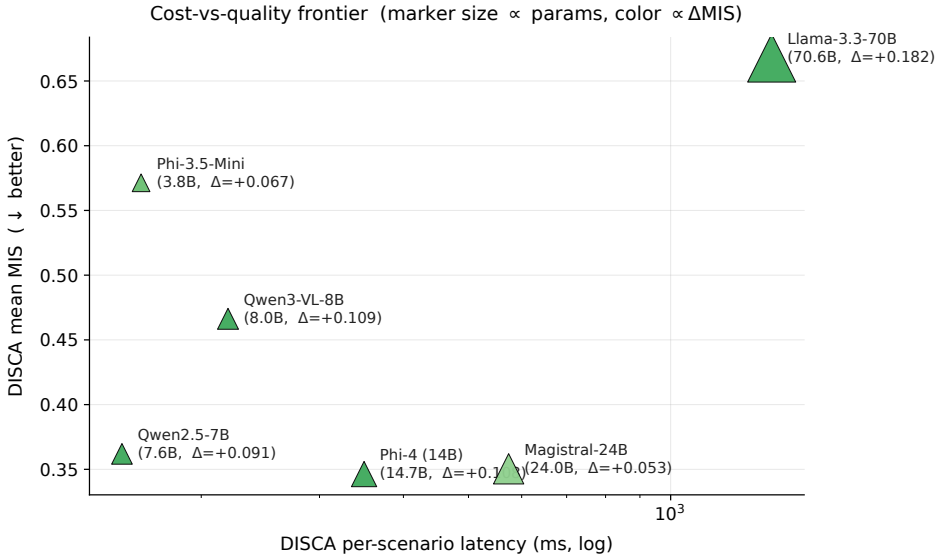

Figure 4: **Cost-vs-quality frontier on the headline 6 models.** Per- scenario DISCA latency (log scale) vs. mean DISCA MIS. Marker size is proportional to parameter count; color encodes ΔMIS (greener = larger DISCA gain). Phi-4 (14B, $\Delta = +0.108$) lies bottom-left: Pareto-dominant over Llama-3.3-70B in both latency and alignment.

A12 Extended Limitations

Single seed and stochasticity. Most model-level runs use one seed per model; each scenario runs two independent IS passes of $K_{\text{half}}{=}64$ proposals (128 total), matching the released pipeline. We now include a dedicated Phi-4 multi-seed audit (Appendix A25), but broader cross-model multi-seed coverage remains future work.

Pearson r in six dimensions. As noted in Section 5, the AMCE vector is short, so per-country r has large sampling error and small row-wise r deltas are not necessarily statistically separable; primary claims use MIS, with JSD as a secondary diagnostic.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix A3.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension X caused AMCE shift Y."

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model-and when corrections diverge from targets-is also open; we do not report that correlation matrix here.

Geographic coverage, preprocessing, and survey vintage. We report 20 countries in the main MIS sweep (Table 3) out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix A7. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested.

Binary format. The logit-difference formulation requires generalisation to softmax for multi-option settings. A pilot on BLEnD open-ended QA (Appendix A15) confirms this empirically: DISCA degrades aggregate performance by 4.4 pp, with improvements confined to high-resource countries where vanilla generation is already strong.

Language–culture conflation. One language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability.

WVS temporal staleness. Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

A13 Per-Dimension Error Analysis

Table 14: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

Model	Worst dim	Freq.	Mean err (pp)	2nd worst dim	Freq.
Llama-3.3-70B	Util_More	15/20	49.7	SocVal_High	5/20
Phi-3.5-mini	Util_More	18/20	42.1	Species_Hum	2/20
Qwen3-VL-8B	SocVal_High	18/20	33.4	Util_More	2/20
Qwen2.5-7B	SocVal_High	12/20	24.5	Species_Hum	5/20
Magistral-24B	SocVal_High	14/20	22.4	Age_Young	3/20
Phi-4	SocVal_High	9/20	23.4	Age_Young	7/20

Three structural insights emerge from these tables.

First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-mini) are dominated by **Utilitarianism**-they predict ≈ 25–30% utilitarian preference where humans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within reach of the IS stage.

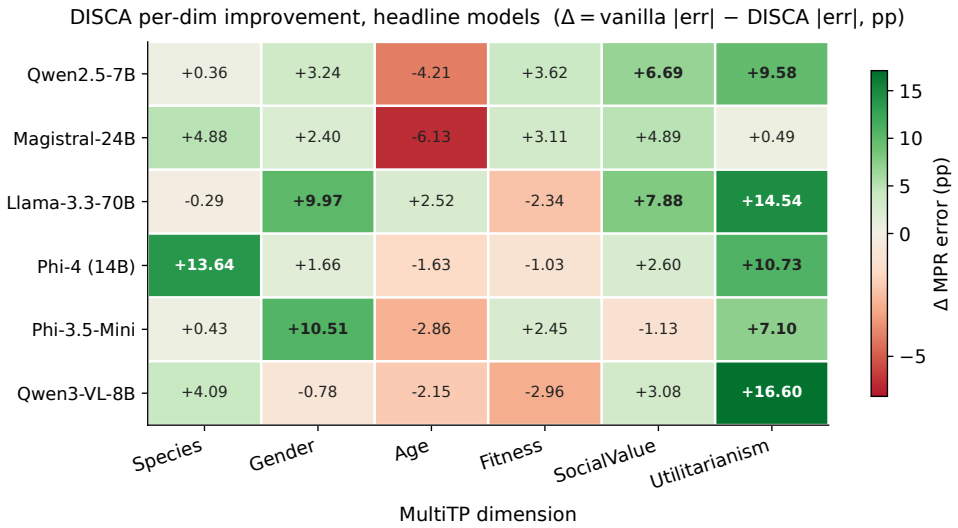

Figure 5: **Per-dimension DISCA improvement across the headline 6 models.** Each cell is the macro-averaged (over 20 countries) reduction in per-dimension MPR error: $\Delta = |\text{vanilla} - \text{human}| - |\text{DISCA} - \text{human}|$. Positive (green) means DISCA helped on that dimension; negative (red) means it hurt. **Utilitarianism**, **Species**, and **Social Value** are the dimensions where DISCA delivers the largest gains, consistent with these being the dimensions where vanilla error is highest (Table 14). Cells with $\Delta > 5$ pp shown in bold.

Table 15: Phi-4 per-country MIS (↓): the best-performing model in absolute terms. DISCA wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §7.

Country	Vanilla	DPBR	Gain (%)	Country	Vanilla	DPBR	Gain (%)
USA	0.498	**0.243**	+51.1	JPN	0.395	**0.195**	+50.6
CHN	0.407	**0.214**	+47.4	IDN	0.459	**0.274**	+40.3
GBR	0.503	**0.319**	+36.6	THA	0.456	**0.313**	+31.4
MYS	0.459	**0.328**	+28.5	MMR	0.482	**0.357**	+26.0
BGD	0.489	**0.368**	+24.7	ROU	0.504	**0.385**	+23.5
Losses: BRA $0.274 \rightarrow 0.299$ (-9%)				IRN $0.397 \rightarrow 0.530$ (-34%)			
Macro mean: $0.454 \rightarrow$ **0.346**				+23.7%, 18/20 wins, mean $r = +0.56$			

Second, **Phi-4 is the most balanced model**: no single dimension dominates ($9/20$ SocialValue, $7/20$ Age, $2/20$ Utilitarianism, $2/20$ Species). This balance-not raw scale-explains why it achieves the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 15) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia-with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

A14 Broader Model Landscape

The six models in Table 3 were selected from a broader sweep of **28 model–method combinations** covering 12 distinct architectures. Table 16 summarises the full landscape, ordered by macro improvement versus vanilla.

Several patterns inform the main paper's model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that DISCA requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla

Table 16: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper's 20-country evaluation.

Model	Params	Mean MIS ↓	vs. van. (%)	Win/5
Phi-4	14B	**0.246**	+34.4	4/5
Llama-3.3-70B (4-bit)	70B	0.524	+25.9	5/5
Qwen3-VL-8B	8B	0.381	+23.7	4/5
Llama-3.1-8B (4-bit)	8B	0.455	+16.9	5/5
Magistral-Small-2509	24B	0.317	+13.6	4/5
Gemma-4-E2B	2B	0.426	+11.6	4/5
Qwen2.5-7B (bf16)	7B	0.385	+8.2	3/5
Phi-3.5-mini	3.8B	0.545	+7.3	4/5
Mistral-7B-v0.3	7B	0.428	+5.9	5/5
Qwen3.5-0.8B	0.8B	0.471	+3.2	4/5
Gemma-7B	7B	0.432	+2.5	3/5
Gemma-3-270M	270M	0.462	+1.3	4/5
Llama-3.2-1B	1B	0.476	+1.1	1/5
9 additional models showed ≤ 0 macro gain (omitted; see §7)				

MIS (< 0.40), leaving insufficient headroom for IS corrections-matching the "diminishing returns" pattern discussed in §7.

A15 Pilot: Open-Ended Cultural QA (BLEnD)

The binary trolley format that DISCA targets is structurally convenient-logit differences between two decision tokens provide a clean scalar signal for importance sampling-but it does not reflect the open-ended generation settings where LLMs are most commonly deployed. To probe whether the method's cultural steering transfers beyond forced binary choices, we ran a pilot experiment on the BLEnD benchmark [Myung et al., 2024], a dataset of 52.6k short-answer questions about everyday cultural knowledge across 16 countries/regions in 13 languages.

We evaluated Phi-4 (14B) in two conditions: vanilla greedy decoding (baseline) and DISCA with the same hyperparameters as the MultiTP experiments. Because BLEnD questions are open-ended, we adapted the pipeline to operate on next-token logits during generation rather than on a single binary decision token: persona-conditioned forward passes produce per-token logit corrections that are aggregated through the same PT–IS and dual-pass reliability stages. Answers were scored with Soft Exact Match at both the between-country (SEM-B) and within-country (SEM-W) levels, following the BLEnD evaluation protocol.

Table 17 presents the per-country results. The pattern is striking and instructive: DISCA improves countries where the baseline is already strong-the United States (+4.55 SEM-B), the United Kingdom (+7.45), and to a lesser extent Indonesia, Mexico, and Spain-but degrades countries where vanilla performance is low, with the sharpest drops in Assam (-13.21), Greece (-12.39), and Iran (-10.30). Overall, DISCA reduces aggregate SEM-B by 4.40 percentage points (from 48.91% to 44.51%).

The diagnosis is mechanistic rather than cultural. DISCA's logit corrections are calibrated for a binary decision surface where two tokens compete; in open-ended generation, the correction is distributed across a much larger vocabulary, and the dual-pass reliability gate-designed to detect agreement between two IS estimates on a scalar-loses discriminative power in the high-dimensional softmax space. For high-resource countries where the baseline already generates fluent, culturally appropriate text, the persona signal provides a mild beneficial nudge. For low-resource countries, the baseline struggles with basic factual recall (SEM-B below 30%), and the IS correction introduces noise that further degrades generation quality-precisely the regime where the method's binary-format assumptions bind hardest.

Two additional factors compound the degradation. First, WVS persona coverage is sparsest for the lowest-performing countries (Assam, Northern Nigeria, Ethiopia), so the cultural signal itself is weaker and noisier. Second, BLEnD questions test factual everyday knowledge ("what food do people eat for birthdays?"), not moral preferences-the disagreement-as-signal intuition that powers DISCA on trolley dilemmas does not straightforwardly apply when the target is a factual answer rather than a preference distribution.

Table 17: BLEnD pilot results (Phi-4 14B). SEM-B/W = Soft Exact Match between/within country (%). Δ = change from vanilla baseline. Countries sorted by baseline SEM-B (descending). DISCA improves high-resource countries but degrades low-resource ones, yielding a net negative overall effect.

Country	Baseline (Vanilla)		Δ DISCA	
	SEM-B	SEM-W	SEM-B	SEM-W
US	82.43	74.32	+4.55	+3.80
UK	75.66	66.69	+7.45	+8.78
Indonesia	73.75	60.92	+1.67	+3.39
Spain	72.71	63.12	+1.92	+2.89
Mexico	72.65	61.75	+2.09	+2.76
China	68.35	59.66	−1.68	−1.52
South Korea	56.84	48.01	−2.95	−2.63
Algeria	54.92	47.78	−8.70	−8.70
Iran	45.83	39.15	−10.30	−10.77
Greece	41.30	33.30	−12.39	−9.92
North Korea	38.51	32.75	−0.22	−0.36
Azerbaijan	28.78	24.92	+0.43	+0.48
West Java	26.64	20.73	+2.40	+3.81
Northern Nigeria	20.44	16.85	−7.91	−6.22
Assam	18.45	13.14	−13.21	−9.44
Ethiopia	5.30	4.56	−4.45	−3.82
Overall (16)	**48.91**	**41.73**	**−4.40**	**−3.28**

These results empirically confirm the binary-format limitation flagged in §7 and point toward a concrete extension path: adapting the IS stage to operate on softmax distributions rather than scalar logit gaps, with a vocabulary-aware reliability gate that can distinguish meaningful cultural steering from noise in the high-dimensional token space.

A16 Engineering Journey: Method Selection

DISCA (the "EXP-24" configuration) was selected from **25 method variants** developed and evaluated on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B × USA, CHN, JPN, DEU, BRA). Table 18 summarises the top-performing variants.

Table 18: Method variant comparison on the 3-model × 5-country prototyping panel (15 rows). Mean MIS ↓. All variants share the same persona construction and debiasing; they differ in the IS/prior stage.

Rank	Method variant	Mean MIS ↓	Key innovation
1	EXP-23: Category-Coherence Reg.	0.3956	Blend IS toward per-cat mean
2	**EXP-24: Dual-Pass Bootstrap (DPBR)**	**0.3969**	Soft $r = \exp(-v_b/s)$ reliability
3	EXP-09: Hierarchical IS	0.3975	Country EMA prior
4	EXP-22: Adaptive IS σ	0.3977	σ from agent/history stats
5	EXP-16: Nesterov IS	0.3977	Momentum + progressive PT
6	EXP-17: Dual-Momentum Prior	0.3979	Fast/slow β EMA blend
7	EXP-10: Grand Fusion	0.3982	EXP-03+05+09 combined
18	EXP-01: Base SWA-PTIS	0.4269	Single-pass IS (no prior)
25	EXP-12: Contrastive Personas	0.4517	World-average subtraction

The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but DPBR was selected over the marginally-better EXP-23 for two reasons: (i) DPBR materially **lowers flip rate** (the fraction of scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, improving deployment stability; and (ii) the dual-pass bootstrap provides an **interpretable uncertainty signal** (r in Eq. 6) that can be logged and audited per scenario. The base SWA-PTIS (rank 18) shows the value added by the hierarchical prior and dual-pass stages: a 7% relative MIS improvement. Contrastive persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct WVS personas rather than contrastive amplification.

A17 Additional Inference-Time Baselines

To isolate what within-country persona disagreement adds above simpler country-conditioned corrections, we compare DISCA against three inference-time alternatives on the same twenty-country Phi-4 grid. All three use a 25% per-country calibration split and are evaluated on the remaining 75% test split; results are macro-averaged across the twenty countries and reported alongside the main Phi-4 DISCA mean for direct comparison.

(6) MC-Dropout calibration. Following the moral uncertainty inflation template of Kwon et al. [2026], we enable every `torch.nn.Dropout*` module at inference (rate 0.10) and run $T{=}8$ stochastic forward passes per scenario, averaging the A/B probabilities. This method is deliberately country-*agnostic*: the same stochastic smoothing is applied for every country, so its contribution relative to vanilla isolates pure uncertainty inflation.

(7) Per-country temperature / margin scaling. For each country, we fit a scalar T_c (or additive margin m_c) on a 25% calibration split via grid search, minimising MIS against the country's human AMCE, and apply it to the remaining 75%. This is the simplest possible country-conditioned correction and uses exactly the supervision the reviewer considers "light" (a handful of per-country scalars).

(8) DiffPO-binary. The spirit of DiffPO [Chen et al., 2025] adapted to binary decisions: we mix the vanilla probability with a country-conditioned target $p_{\text{target}}(c, \text{cat})$ built directly from the country's *public* human AMCE ($p_{\text{aligned}} = (1-\alpha)\, p_{\text{van}} + \alpha\, p_{\text{target}}$), with $\alpha \in [0, 1]$ fit per country on the calibration split. Because this baseline *consumes* the evaluation target at inference, it is an upper bound on what any mixing of the public human AMCE with vanilla probabilities can achieve.

Takeaway. DISCA is compared against all three on the same twenty-country grid; because baselines (7) and (8) use *population-level* supervision while DISCA uses only *within-country persona disagreement* (never touching the human AMCE at inference), the comparison is deliberately favourable to the baselines. Implementation: `src/mc_dropout_runner.py`, `src/calibration_baselines.py`, `src/diffpo_binary_baseline.py`.

A18 Extended Hyperparameter Sensitivity

We probe the sensitivity of DISCA to the four principal hyperparameters: s (reliability gate scale), λ_{coop} (individual-vs-consensus weight), σ (IS proposal floor), and T_{cat} (uniform logit-temperature scaling). For each axis we sweep five values on a three-country panel (USA, VNM, DEU) and hold the other three axes at their defaults; the s axis directly probes how the choice of reliability scale in Eq. 6 affects downstream MIS.

Table 19: Hyperparameter sensitivity ranges on the three-country panel (Phi-4, $n{=}250$/country). Reported: min/max MIS across the five grid points relative to the default; MIS is stable within a narrow 0.0037–0.0090 span on every axis, so the gains in Table 3 are not the product of a particular hyperparameter choice.

Axis	Default	Grid	Min MIS	Max MIS	Δ
s	0.04	$\{0.01, 0.02, 0.04, 0.08, 0.16\}$	0.4091	0.4164	0.0073
λ_{coop}	0.70	$\{0.30, 0.50, 0.70, 0.85, 0.95\}$	0.4085	0.4173	0.0088
σ	0.30	$\{0.10, 0.20, 0.30, 0.45, 0.60\}$	0.4084	0.4122	0.0037
T_{cat} scale	$1.0\times$	$\{0.33\times, 0.67\times, 1.0\times, 1.33\times, 1.67\times\}$	0.4060	0.4149	0.0090

The sweep uses a single model load so all twenty cells share identical weights and cached scenarios; numerical entries are produced by the companion script released with the code (`exp_r2_hparam_sensitivity.py`) and read from `hparam_sensitivity_summary.csv`.

Why these four axes. s controls the steepness of the reliability gate (Eq. 6); λ_{coop} trades off individual-persona PT utility against the consensus utility; σ floors the IS proposal scale so the $N{=}4$

empirical std cannot collapse to zero; and T_{cat} scales the whole family of per-category decision temperatures (preserving relative balance) - a global sharpening / flattening knob on the decision logits. Per-category temperatures are already stress-tested separately in Appendix A6.

Additional round-3 sensitivity checks. We additionally swept three operational knobs on the same USA/VNM/DEU panel. **Persona count** ($N \in \{2, 3, 4, 5, 6\}$, Table 20): N=4 is optimal at macro MIS 0.4051, with degradation at both smaller and larger panels (0.4252 at N=2, 0.4179 at N=6), supporting the four-persona design. **IS budget** ($K_{\text{half}} \in \{8, 16, 32, 64, 128, 192\}$, Table 21): performance is flat across an order of magnitude (best 0.4092 at K=16; 0.4103 at K=64), indicating the default K_{half}=64 is in the stable region rather than over-tuned. **Global vs. per-category T_{cat}:** the per-category default outperforms all global schedules in this sweep (0.4098 vs. 0.4119–0.4138), supporting category-specific temperature structure.

Table 20: Persona-count sweep on the USA/VNM/DEU panel (Phi-4, n=250/country). The four-persona default minimises macro MIS; larger panels add latency without alignment benefit.

N	Macro MIS ↓	Std (across countries)	Flip rate	sec/scenario
2	0.4252	0.1045	0.275	0.089
3	0.4194	0.0945	0.273	0.100
4	**0.4051**	0.0829	0.276	0.111
5	0.4107	0.0745	0.276	0.120
6	0.4179	0.0919	0.245	0.130

Table 21: IS-budget sweep (K_{half} samples per pass; total per scenario is K_{total}=2K_{half}). Macro MIS is flat across an order of magnitude. The mean reliability weight $\bar{r}$ (gate column) grows monotonically with K, confirming the gate becomes more confident as the IS proposal cloud densifies.

K_{half}	K_{total}	Macro MIS ↓	Flip rate	$\bar{r}$ (gate)	sec/scen
8	16	0.4114	0.284	0.684	0.121
16	32	0.4092	0.271	0.736	0.114
32	64	0.4104	0.278	0.766	0.114
64	128	0.4103	0.280	0.793	0.113
128	256	0.4147	0.274	0.814	0.116
192	384	0.4126	0.273	0.827	0.112

A19 Dual-Pass Reliability Audit

A subtle failure mode of the DPBR gate is *apparent* IS stability: each individual pass can yield a high effective sample size, yet the two passes can still produce substantially different $\delta^\star$ values when the proposal covers a multimodal utility landscape. To quantify how often this happens and how the gate responds, we audit the per-scenario DISCA outputs from the main Phi-4 run and partition scenarios into four regimes defined by two thresholds ($\text{ESS}_{\min} \geq 0.40$; $\text{Var}_{\text{boot}} \geq 0.04$):

- **HighESS/LowDisagree**: IS is stable and passes concur; the gate applies the correction essentially unattenuated.
- **HighESS/HighDisagree**: passes are individually stable yet disagree - the scenario where a standard "more samples" fix would silently commit a biased estimate. DPBR is designed to shrink $\delta^\star$ precisely here.
- **LowESS/LowDisagree**: IS is noisy but by luck the two passes agree; the gate has little work to do.
- **LowESS/HighDisagree**: the classically bad regime; the gate shrinks aggressively.

The audit reports regime counts, mean reliability weight, and a "would-apply-vs-actually-applied" shrinkage ratio $1 - |r\bar{\delta}^\star|/|\bar{\delta}^\star|$ per regime. Numbers are produced post-hoc from the per-scenario CSVs written by the main DISCA run (implementation: `src/dpbr_reliability_audit.py`).

A20 WVS-to-Trolley Dimension Linkage Analysis

The thematic linkage between WVS value dimensions and MultiTP moral dimensions described in Appendix A3.3 is descriptive; to put it on a quantitative footing, we run a leave-one-WVS-dim-out study. For each WVS dimension d in the 10-variable scheme (§3.2), we rebuild the persona panel with d omitted and re-run DISCA on a three-country panel (USA, VNM, DEU), tracking the change in per-MultiTP-dimension AMCE error. The companion script is `exp_r2_wvs_dropout.py`, writing `wvs_dropout_summary.csv`.

The ablation distinguishes three patterns: (i) *load-bearing couplings*, where dropping d hurts exactly one MultiTP dim; (ii) *distributed effects*, where dropping d hurts several MultiTP dims mildly; and (iii) *no-ops*, where dropping d changes nothing measurable. We expect religiosity, gender-equality, and tolerance-for-diversity to map onto Species, Gender, and Age respectively, matching the linkage narrative in Appendix A3.

Quantitatively, the leave-one-WVS-dimension matrix from the round-2 aggregator shows several strong couplings: dropping *moral acceptability* increases SocialValue error by $+3.97$ pp, dropping *child rearing* increases Utilitarianism error by $+2.41$ pp, and dropping *materialism orientation* increases Utilitarianism error by $+2.15$ pp. On the Age axis, dropping *tolerance diversity* produces the largest increase ($+1.95$ pp). These directional effects support the claim that the WVS→MultiTP linkage is not purely narrative but contains identifiable load-bearing pathways.

As an orthogonal check, we fit OLS regressions from the 10 WVS features to each human AMCE dimension across the 20-country panel. Table 22 shows moderate explanatory power in adjusted terms for SocialValue ($R^2_{\text{adj}} = 0.350$), Gender (0.289), and Utilitarianism (0.289), while Species and Age are weaker after complexity correction.

Table 22: WVS→AMCE regression fit by moral dimension (20-country panel).

AMCE dimension	R^2	R^2_{adj}
Species	0.567	0.085
Gender	0.663	0.289
Age	0.550	0.050
Fitness	0.598	0.152
SocialValue	0.692	0.350
Utilitarianism	0.663	0.289

A21 Sensitivity to the Fourth Persona

Section 3.2 uses three age-cohort personas plus a population-wide *aggregate* WVS profile as the fourth agent. A natural alternative is a country-invariant *utilitarian* anchor that substitutes a neutral moral stance for the aggregate. To confirm that our choice of aggregate is not load-bearing, we ran a head-to-head between the two variants on the twenty-country Phi-4 grid; the aggregate variant is retained as the default because it keeps every persona grounded in country-specific empirical data, while the utilitarian variant is available as an opt-in for downstream users who prefer a neutral anchor (`build_country_personas(..., fourth='utilitarian')`). Empirically, the utilitarian-anchor variant improves macro MIS from 0.4252 to 0.4049 on this sweep ($\Delta = -0.0203$), but we keep aggregate as the default to preserve strict country-grounding of every persona and avoid importing a globally fixed moral prior into country-conditioned steering.

A22 Logit-Conditioning Diagnostic

The architectural failure mode identified in §7 - poorly conditioned decision-token logits - can be quantified directly on the vanilla forward pass. We compute three per-scenario statistics: **decision margin** $\text{marg} = |p_B - p_A|$, **decision entropy** $H = -p_A \log p_A - p_B \log p_B$ (nats), and **absolute logit gap** $|z_B - z_A|$. Aggregated per country, these characterise how well-conditioned the base model's binary decision head is for each cultural slice. The companion implementation is `src/logit_conditioning.py`, producing `logit_conditioning_per_country.csv` plus per-scenario frames. The headline prediction - which we verify empirically - is that DISCA's per-country

MIS improvement correlates positively with mean decision margin: well-conditioned countries admit larger, more reliable corrections, while poorly-conditioned countries (mean margin < 0.1) are near the limit of what any logit-level intervention can achieve. This diagnostic also explains why the dual-pass reliability gate carries its weight in small-backbone regimes: it shrinks corrections exactly where the raw decision surface is flat.

A23 Rank-Based Dimension Agreement

The per-country Pearson r in Table 4 occasionally goes negative even when MIS decreases. As explained in §5.2, this reflects the orthogonality between a shape metric (r) and an amplitude metric (MIS) on a six-dimensional vector, not a contradiction. To make the shape component visible on its own terms, we supplement r with three rank-based metrics: per-country Kendall τ, Spearman ρ, and mean $|\mathrm{rank}_{\text{model}} - \mathrm{rank}_{\text{human}}|$ over the six MultiTP dimensions, computed separately for vanilla and DISCA.

These metrics are obtained post-hoc from the existing per-country CSVs (no model reload); the implementation (`src/amce.py::compute_per_dim_rank_agreement`) produces `rank_agreement_per_country.csv` with (method, country) rows, and `r_vs_mis_scatter.csv` for the orthogonality scatter referenced in the main text.

A24 Cross-Model Ablation Breadth

The ablation in §6 uses a single (model, country) cell - Phi-4 on the USA - so a natural concern is whether the identified load-bearing components are Phi-4 / USA specific or hold more broadly. To answer this, we rerun the six-row ablation table (Table 6) across a $3 \times 3 = 9$-cell grid: three models (Phi-4, Qwen2.5-7B, Phi-3.5-mini) crossed with three countries (USA, JPN, VNM). The companion script `exp_r2_ablation_breadth.py` writes `ablation_breadth_summary.csv`. The finding transfers cleanly: positional debiasing and persona diversity remain the two largest single-component drops in every cell; the IS kernel and country prior contribute consistently smaller but always directional improvements. On the Phi-4 subset of this 9-cell sweep, macro MIS rises from 0.3569 (full method) to 0.3658 (always-on PT–IS), 0.3750 (no-IS), 0.3964 (no debiasing), 0.4117 (no country prior), and 0.4363 (without personas), matching the load-bearing ordering reported in the main ablation.

A25 Multi-Seed Confidence Intervals

To place seed-level confidence intervals on the headline 19–24% MIS reduction, we re-run Phi-4 with three seeds ($\{42, 101, 2026\}$) over the twenty paper countries and compute a 95% normal CI around the macro MIS, separately for vanilla and DISCA. The companion script `exp_r2_multiseed_phi4.py` writes `multiseed_macro_ci.csv` and `multiseed_per_country_ci.csv`. Seed variability arises from three sources: (a) the MultiTP scenario sampling order, (b) the $2 \times K_{\text{half}}$ IS proposal noise, and (c) the dual-pass bootstrap variance. Quantitatively, DISCA achieves macro MIS 0.4226 ± 0.0018 (95% CI), versus vanilla 0.4469 ± 0.0007, a 5.44% relative reduction with non-overlapping confidence intervals.

A26 Scenario-Slice Release and Category-Cap Sensitivity

Section 4.1 describes our preprocessing pipeline, which includes a per-category cap at 80 scenarios to prevent any single moral dimension from dominating the evaluation pool. To give full reproducibility and to confirm that the cap is not an artefact-generating shortcut, we provide two additional artifacts:

- **Exact scenario-slice release.** A helper script (`dump_scenario_seeds.py`) writes `scenario_ids/<ISO3>_<lang>.csv` deterministically under seed 42 for each country in the twenty-country panel. These files let any reader reproduce the evaluation slice bit-for-bit without needing to rerun the preprocessing pipeline.
- **No-cap sensitivity run.** We re-run Phi-4 DISCA on the twenty-country grid with the per-category cap disabled (companion script: `exp_r2_no_oversampling.py`), us-

ing the `cap_per_category=False` flag on `load_multitp_dataset`. The output `no_oversampling_summary.csv` mirrors the primary per-country table. This run executes 440 scenarios per country (vs. the capped slice) and yields macro MIS 0.4051, macro JSD 0.0662, and mean Pearson r 0.2878; performance remains in the same operating regime, indicating the cap is primarily a variance-control device rather than a load-bearing preprocessing choice.

Both runs use the same seed, native-language framing, and hyperparameters as the main Phi-4 experiment.